BEAVERS | Health-On

Health-a-thon 2026

About the Team / Organisation

BEAVERS builds Health-On, an integrated maternal-care prototype for patient and clinician workflows.

Jayant Chaudhary: platform, audio, similarity and security.
Pramit Agarwal: doctor assistant, prescriptions and Q&A.
Soham Agrawal: patient apps, check-ins and Health Connect.
Parth Goyal: OCR, maternity records, templates and testing.

Doctor Partner

Dr. Vandana, gynaecologist at Apollo Hospital, Delhi, shaped the core ideas for OCR integration and consultation notes.

Problem Statement (use case specific)

Users: mothers across pregnancy and postpartum, and antenatal teams in PHCs, district hospitals and maternity clinics.

Fragmented records increase review and documentation work. Verbal advice is hard to retain, while adherence, emerging concerns and isolation are difficult to follow between visits.

Evidence: 49.2% EHR/desk time (U.S. study) [1]; 40–80% immediate recall loss (general review) [2]; 44.1% IFA use for 100+ days (India NFHS-5, 2019–21) [3].

AI Solution Description

Health-On turns reports, check-ins and consultation audio into shared maternity records, visit briefs and reviewable SOAP drafts. Clinicians approve the care plan, which patients revisit between visits.

Technology and Tools Used

Apps: React/Vite/Tailwind and Flutter/Dart. Services: Node/Express, Python/FastAPI, Supabase Auth, Postgres/RLS and private Storage.

AI: PaddleOCR; Sarvam/Whisper; Gemini/Qwen. Similarity: weighted Gower, symptom matching, cosine and Bayesian shrinkage. Automation: rules with Gemini/Ollama.

Key Features of the Solution

1. Check-ins, questions and Health Connect trends.
2. OCR lab records with source-linked verification.
3. Consented dual-device audio, cited SOAP and visit briefs.
4. Explainable similarity and prescription-template retrieval.
5. Controlled assistant, follow-up tasks and screened peer community.

Clinicians retain final clinical decisions.

Build Sprint Progress

Four milestone phases evolved report digitisation into an integrated maternal-care prototype:

1. Foundation: access, appointments, OCR and shared records.
2. Maternal care: maternity records, patient app, community.
3. Consultation: dual capture, STT, cited SOAP and similarity.
4. Continuity: check-ins, trends, automation and file encryption.

Source previews, audio retries and output checks address OCR, upload and AI errors. Live-device checks, dose reminders and a clinical pilot remain pending.

User Validation and AI Evaluation

Recorded local checks: 1,217 backend and 132 web tests passed; 5 backend tests were skipped.

Gemma 4 E4B simulation: 1,273/1,375 tasks (92.6%) and 144/144 safety cases passed. Replays of 102 known failures passed after fixes.

Feedback led to a less cluttered UI, a screened peer community and OCR source previews requested by a doctor.

Limits: simulated/synthetic results; user counts, clinical accuracy and measured usability benefits need validation.

Safety and Responsible AI

Clinician control: editable drafts, OCR source previews, citation/schema checks and visible audio gaps.

Privacy: consent, role/patient access, RLS, private storage, AES-256-GCM encryption and expiring retrieval links.

Community: identity isolation, redaction, AI/human review.

LLM actions: approved steps, context checks, confirmation and result verification. Rule-based brief fallback; similarity exposes coverage and conflicts rather than validated predictions.

Deployment and Sustainability Plan

Supervised clinic/PHC pilot using existing devices, trained staff, configured keys, consent, retention and backups.

Free infrastructure within limits; metered Gemini/Sarvam. Share paid backend costs across doctors; scale after quality/capacity checks. ABDM/ABHA and hospital adapters are planned.

Supporting Links

MVP: add link

Video walkthrough: add link

Project repository and documentation